

Practical Data Science I

a.k.a. Wrangling Data and Answering Questions

Adriane Fresh

What is Data Science?

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2. What (**empirically**) **is** Data Science?

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Study of how to solve problems by answering questions using quantitative data.

- Problem-first approach.
 - Data Science is an *applied* discipline!
- The tool you use should be dictated by the question you want to answer. (Not tools for tools' sake.)

What (empirically) is Data Science?

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Over the past **several decades**:

1. Availability of data ↑
2. Computational power ↑

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⇒ Huge proliferation and increase in sophistication of computational methods.

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 - Statistics
 - Economics
 - Political science
 - Engineering
 - Public Policy
 - Biostatistics

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⇒ Development of new tools occurred **within each silo**.

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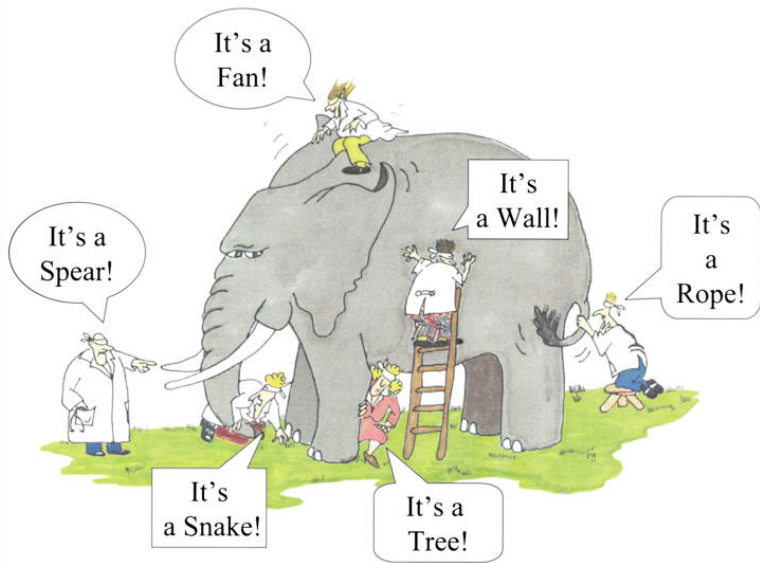
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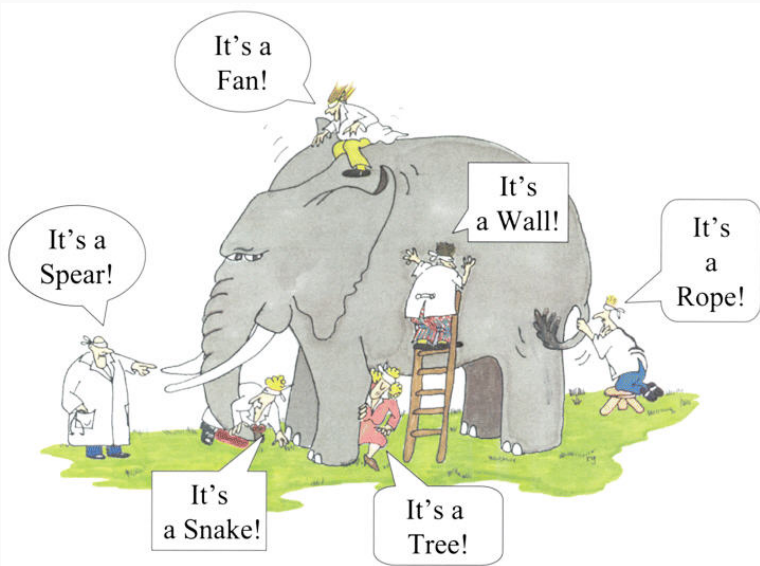
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- Each silo has focused on the aspects **most relevant** to their applications. e.g.:

Where are we today?

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- Every silo has its **own vocabulary**.
- Each silo has focused on the aspects **most relevant** to their applications. e.g.:
 - **CS** likes to classify things and make predictions; don't care how the model works.
 - **Social scientists** like to make causal claims; don't care (as much) about predictive power.





⇒ This is where we are *now*.

What is (empirically) Data Science?

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An effort to unify the development of quantitative methods.

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An effort to **unify the development of quantitative methods**.

→ *Recognize the elephant.*

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In many ways, *you will have better perspective than your professors.*
- Expect fragmentation in both software and vocabulary.
The fact things don't always seem coherent isn't because you're missing something.

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- My people who talk about data science were trained in a single silo.
- And may not fully recognize what's happening beyond.
 - That's why **this course** was designed by faculty **from many fields, for people from many fields.**

This Class

“The Python Class”

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- *Data Science* in Python

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- *Data Science* in Python
 - Python standard library, numpy, pandas, scikit-learn, matplotlib, statsmodels

“The Everything-That-Comes-Before-Data-Analysis Class”

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In this course, you will learn:

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All through hands-on experience!

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⇒ Emphasis is on **generalizable data science skills** throughout.

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- Find and organize data *on your own*.

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By the end of this class:

- Find and organize data *on your own*.
- Understand how to clean, merge, and manipulate real-world data.
- Know how to approach organizing a full project.
- Troubleshoot messy real world data.

Why Python?

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And why not:

- Stata
- R (Tidy-Verse)
- R (Base-R)

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And why not:

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Excellent for tabular data, some text
- R (Tidy-Verse)
Excellent for tabular data
- R (Base-R)
Good for tabular, network, geospatial, some text and ML

?

Why Python?

Python:

- Tabular data
- Network data
- Geospatial data
- Natural Language Processing (NLP)
- Neural Networks
- Using with Cloud Compute
- Big Data
- Large Language Models
- All Machine Learning
- ...

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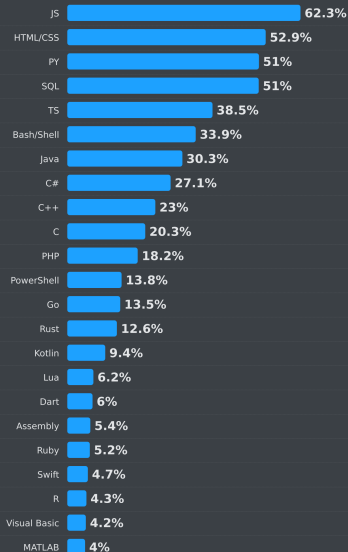
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Network Effects:

- 90s and 2000s (even 2010s): most social scientists used Stata or R, developed for Stata and R.
- 2010s and 2020s: Businesses, computer scientists and software engineers moved to data science.
 - Wanted a fully-featured, general purpose language (ideally one they know).

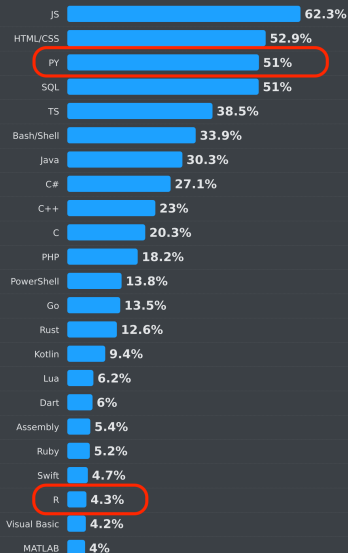
Stack Overflow Developer Survey 2024

Programming, scripting, and markup languages

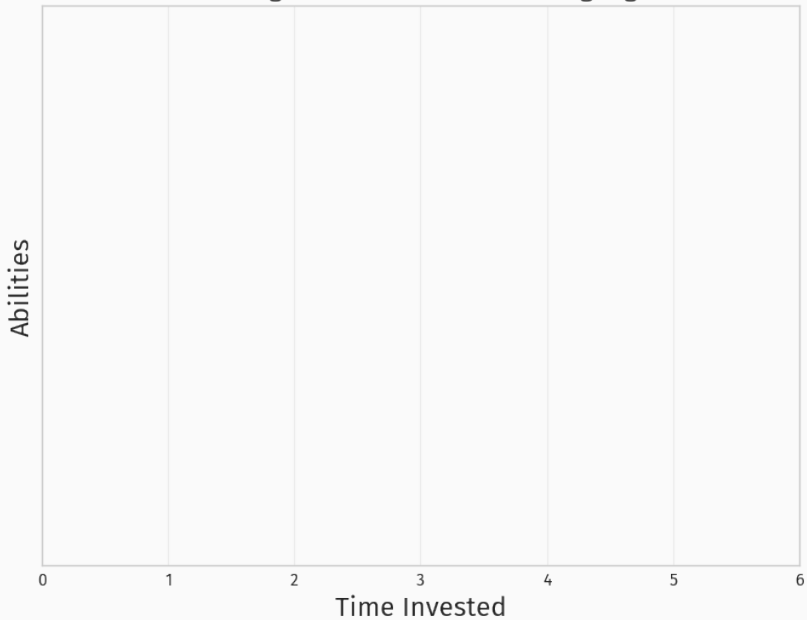


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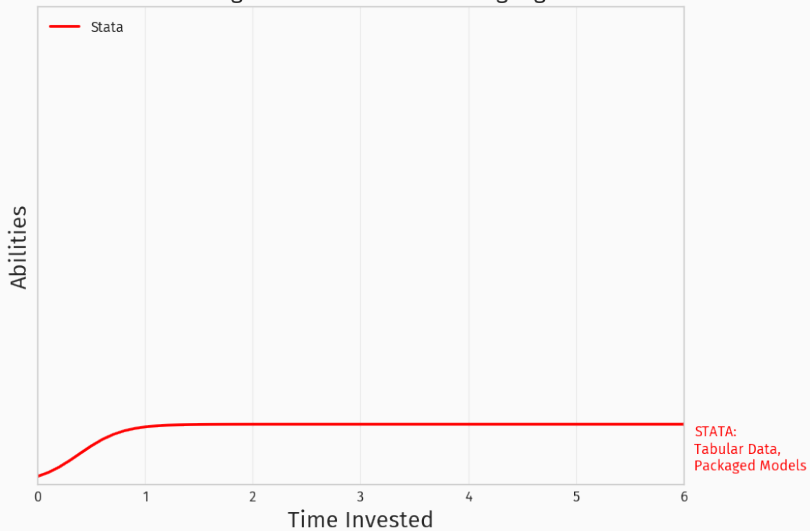
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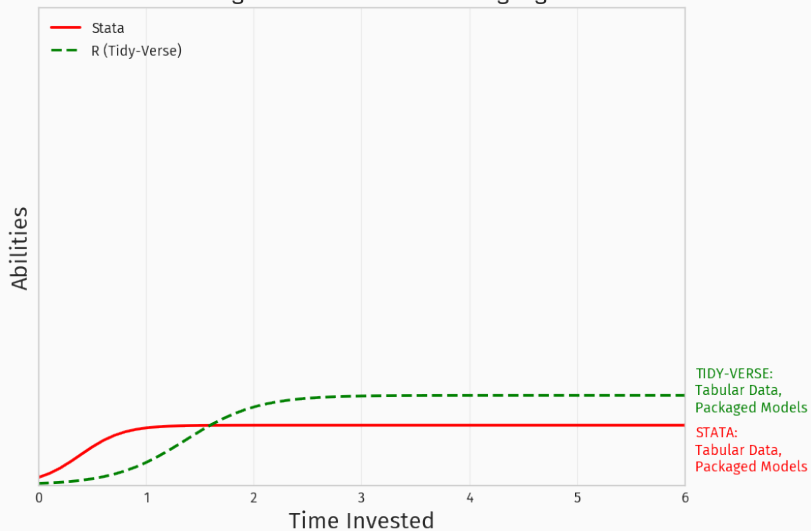
Learning Curves for Different Languages



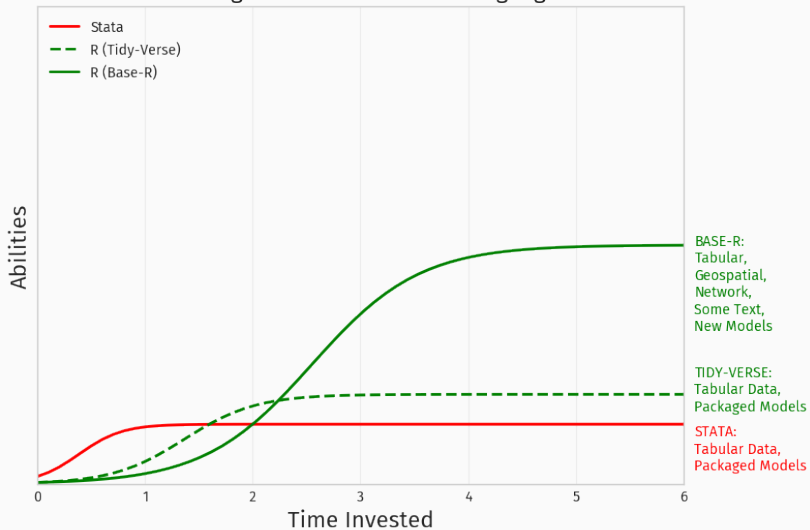
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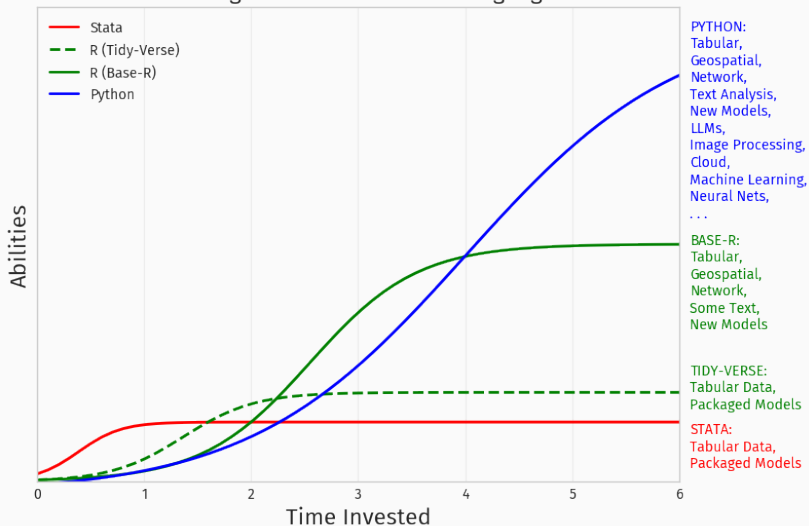
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So what?

- “This is so much easier to do in [Stata/R]”
 - You’re not wrong!
 - (Also easier ways in Python!)
- “This isn’t what I want to learn.”
 - Especially for the first 4 weeks.

By the end of this class...

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- and more.

If you also take IDS 541...

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You will have a strong understanding of computational thinking and Python that will allow your empirical work to go wherever your research takes you, instead of feeling limited by what existing packages make easy.

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- e.g., including the Industrial Revolution in Britain, the enfranchisement of Black people in the U.S., regime change in Chile, and contemporary U.S. election administration. I am particularly interested in causal inference with text, and measurement with historical texts (both handwritten and typed).

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www.practicaldatascience.org